Dynamic Programming on a Grid

Optimal Control, Homework 8

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Problem

The system is $x(k+1) = x(k) + u(k)$, with $0 \le x \le 1.5$, $|u| \le 1$ and $x(0) = 1.5$. The cost is $J = x(N)^2 + \sum_{k=0}^{N-1} 2u(k)^2$, minimised by backward dynamic programming on quantised state and input grids.

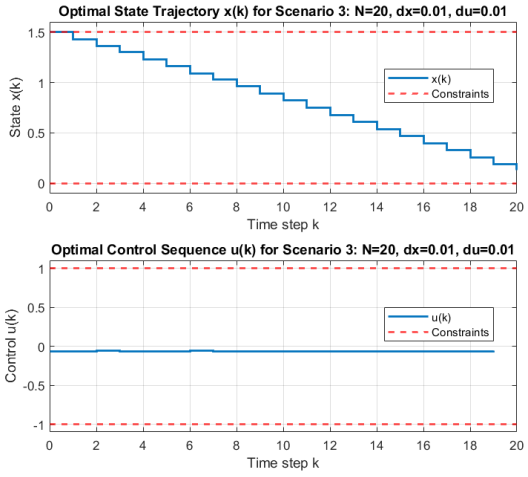
Method (`Q8.m`)

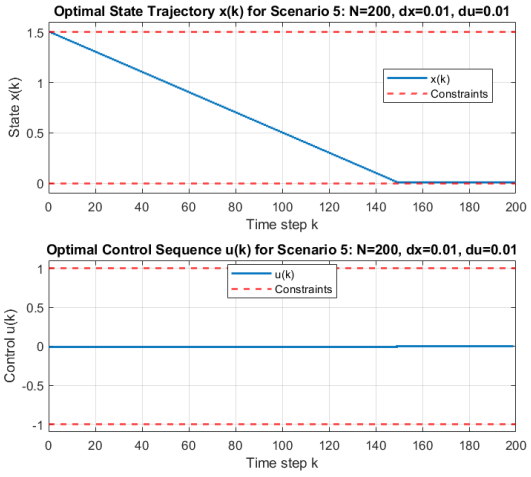
The terminal cost $J_N(x) = x^2$ is stored on the state grid. Going backwards, for every grid state I try every grid input that keeps $x(k+1)$ feasible, interpolate the cost-to-go, and keep the best input. A forward pass from x_0 then recovers the optimal trajectory.

Results (re-run in R2023b)

Scenario	N	grid step ($\Delta x = \Delta u$)	optimal cost J
1	2	0.5	1.2500
2	20	0.1	0.2900
3	20	0.01	0.2051
4	200	0.1	0.2900
5	200	0.01	0.0299

With a longer horizon the controller can spread the effort over more steps and the cost drops. A finer grid lets it use smaller, cheaper inputs. On the coarse grid, going from $N = 20$ to $N = 200$ does not help, because the smallest non-zero input step (0.1) already limits the cost.





Note: the printed label of scenario 1 in the script says "dx=0.01", but the code uses $\Delta x = \Delta u = 0.5$, which is the value in the table.